

Electric Vehicle Scheduling and Vehicle-to-Grid Integration in Microgrids

Nathan Cho¹, Andrea Lodi², and Anna Scaglione³

¹Operations Research and Information Engineering, Cornell Tech, New York, NY, USA
(nc437@cornell.edu)

²Jacobs Technion-Cornell Institute, Cornell Tech and Technion - IIT, New York, NY, USA
(al748@cornell.edu)

³Electrical and Computer Engineering, Cornell Tech, New York, NY, USA
(as337@cornell.edu)

August 19, 2025

Abstract

The logistical challenges and high costs associated with procuring and transporting fuel to remote military bases underscore the need for sustainable and resilient energy solutions. Integrating renewable energy sources and electric vehicles into military microgrids offers a promising approach to enhance energy security and operational readiness. This paper explores the optimization of travel, charging, and discharging schedules for a fleet of military electric trucks, aiming to minimize reliance on fuel-generated electricity while ensuring that mission-critical transportation needs are met. We extend the classical Vehicle Scheduling Problem by incorporating Electric Vehicle Scheduling Problem dynamics and Vehicle-to-Grid capabilities, developing a comprehensive optimization model that addresses both logistical and energy demands within a military microgrid context. Utilizing a column generation approach, we efficiently solve large-scale instances and demonstrate significant improvements in fuel efficiency, renewable energy utilization, and overall operational cost. Computational experiments using realistic demand and solar generation data illustrate that integration of vehicle-to-grid enabled electric vehicles substantially reduces fuel consumption and can generate surplus energy returned to the grid. The results indicate that while battery constraints may require an increased fleet size, strategic scheduling of charging and discharging yields considerable economic and operational benefits. Our findings provide valuable insights for planners aiming to optimize energy use, reduce dependence on traditional fuel sources, and enhance operational resilience in remote environments. **Keywords:** Vehicle routing Electric vehicles Vehicle-to-grid Renewable energy Column generation

<https://arxiv.org/submit/6693871/view>

1 Introduction

Military operations in remote locations face substantial challenges in securing reliable energy supplies due to logistical complexities and high transportation costs. Delivering diesel fuel to isolated bases is particularly costly and susceptible to disruption by adversaries targeting supply lines [11, 30]. These vulnerabilities underscore the critical need for more sustainable and resilient energy solutions in military contexts.

To mitigate these challenges, the U.S. Department of Defense has been exploring the integration of renewable energy sources and electric vehicles (EVs) into military operations. Renewable energy can provide reliable and resilient power in various configurations, improving energy security and reducing dependence on traditional fuel supplies [17]. Moreover, EVs, particularly when equipped with vehicle-to-grid (V2G) capabilities, can serve two functions: meeting transportation requirements and functioning as energy storage systems capable of supporting grid operations by discharging energy during peak demand [3].

However, the intermittent nature of renewable energy requires effective energy storage solutions to ensure a continuous power supply. Sufficient battery storage is crucial to balance supply and demand, maintaining the operational readiness of both the base’s electrical systems and its fleet of EVs [10].

Our research focuses on optimizing the scheduling of electric military trucks to minimize the reliance on fuel-generated electricity within microgrids. Unlike previous studies that predominantly target civilian applications, we specifically address the unique demands and constraints of military operations in isolated environments. By extending the classic Vehicle Scheduling Problem (VSP) to incorporate Electric Vehicle Scheduling Problem (EVSP) dynamics and integrating V2G capabilities, our model comprehensively addresses both logistical scheduling and microgrid energy management.

Prior EVSP literature typically falls into two categories: the first class addresses travel and charging optimization without V2G capabilities, employing exogenous grid prices, whereas the second class exclusively emphasizes V2G demand-response functions, neglecting travel scheduling [25, 15, 24, 7].

In contrast, our integrated optimization framework jointly considers travel scheduling, charging, discharging, and energy management, directly targeting fossil fuel reduction while ensuring that all necessary missions are completed. In addition, we introduce a novel modeling feature that treats standalone battery operations as separate dispatchable schedules, enabling the optimization of mobile and stationary energy assets within the same framework.

We utilize a column generation approach to efficiently solve large-scale instances, demonstrating substantial improvements in fuel efficiency, renewable energy utilization, and overall operational cost-effectiveness. Our computational experiments use realistic electricity demand and solar generation data, confirming that EV and V2G integration significantly enhances energy resilience and economic performance.

This paper is organized as follows. Section 2 reviews the relevant literature on electric vehicle scheduling. Section 3 presents the problem setting and defines the key assumptions that distinguish our approach. Section 4 introduces the classical VSP formulation, serving as the baseline for our extended model. In Section 5, we present our newly developed EVSP-V2G model that solves vehicle scheduling while minimizing electricity generation by allowing EVs to store renewable energy and perform V2G operations to help meet demand. Section 6 evaluates the proposed framework through computational experiments employing realistic solar generation and demand profiles. Finally, Section 7 summarizes key findings and suggests future research directions.

2 Literature review

Early research on vehicle scheduling provided foundational methodologies for multi-depot scheduling, later adapted for electric buses [20]. Studies on mixed bus fleet scheduling integrates both electric and conventional buses to enhance efficiency under range and refueling constraints [16]. The scheduling of electric buses within public transportation networks has also been examined, with considerations for operational constraints and real-world implementation [28]. Perumal et al.

[21] categorize EVSP methodologies, ranging from exact approaches to heuristic and metaheuristic strategies.

Several studies have introduced novel constraints such as capacitated charging stations and partial charging, enhancing solution realism [5]. Robust scheduling strategies have been proposed to mitigate uncertainties due to stochastic traffic conditions [27]. Wu et al. [29] explore a multi-depot EVSP incorporating power grid characteristics, highlighting the interplay between transportation and energy networks. Additionally, mixed-fleet scheduling has been analyzed in terminal-based systems, demonstrating potential improvements in fleet operations [22].

Battery degradation and non-linear charging behaviors significantly influence scheduling decisions. Zhang et al. [31] propose an optimization model considering these factors to extend battery lifespan and improve scheduling efficiency. Desaulniers et al. [6] provides a foundational survey of exact algorithmic techniques for electric vehicle routing problems with time windows, including branch-price-and-cut and labeling algorithms that directly support column generation-based EVSP methods. Exact approaches, such as branch-and-price and integer programming, have been applied to optimize fleet scheduling while accounting for charging limitations [12]. Integrating scheduling with optimal charging strategies has also been studied, offering solutions that jointly optimize routing and energy consumption [23]. Furthermore, exact solution approaches for dispatch problems have been proposed, demonstrating practical applicability in urban transit networks [2].

Few studies also explicitly consider V2G capabilities or microgrid energy interactions within fleet scheduling. Some recent works address the role of EVs as distributed storage in power systems. Sedighizadeh et al. [26] show that incorporating EV charging schedules in a microgrid can reduce operational costs through peak shaving and valley filling. Józwiak et al. [13] demonstrate in an island microgrid setting that controlled bidirectional charging of EVs significantly increases renewable energy utilization and yields up to 50% cost savings. Real-world trials in military settings further indicate the potential of V2G and vehicle-to-vehicle (V2V) systems to significantly improve fuel economy [18].

Contributions Building on the above research, this paper proposes a novel EV scheduling model for an isolated microgrid environment with renewable energy. Unlike existing EVSP literature, our formulation explicitly accommodates bidirectional energy flows, allowing vehicles to charge from local solar surpluses and discharge back to the grid or other vehicles as needed. Through computational experiments using realistic data, our analysis reveals important trade-offs among fleet size, trip coverage, and fuel consumption under varying mission intensities and solar availability. The results highlight the significant potential of optimally scheduled V2G-enabled EVs to dramatically reduce fossil fuel reliance without compromising operational effectiveness. Overall, this study provides valuable insights into enhancing energy security and sustainability for transportation operations within microgrids. Additionally, by relaxing the variables such as trucks, batteries, and generation capacity to be unbounded, the model identifies minimum infrastructure requirements needed to support feasible operations. In doing so, it indirectly informs planning decisions such as the number of electric vehicles and batteries required for deployment.

3 Problem Description and Assumptions

We focus our attention to a specific setting, one based on a real example: a geographically isolated military base (e.g., a base on a remote island or in the desert) equipped with the ability to generate power by using gasoline and solar energy, where a fleet of electric trucks can draw power from the grid and discharge energy back into it during periods of high demand. In addition to managing

charging and discharging schedules, the trucks must carry out a set of predefined tasks (referred to interchangeably as trips in this paper).

In our setting, we make the following assumptions: first, we can forecast both the weather for renewable energy, as well as the electricity demand. Second, there is no limit to the capacity for charging and discharging trucks at once. For the sake of solving the planning problem, we will also assume that there are an unlimited number of trucks available to carry out all tasks. Third, we assume that sufficient generation capacity is always available to simultaneously meet both the electrical demand and the charging requirements of any number of vehicles. This implies that there are no constraints on generation capacity or grid infrastructure, even under high load conditions. We also assume a simple case of linear charging and no degradation of batteries. For the sake of modeling, we will assume we have an unlimited number of trucks that can be used to perform tasks and help meet charging or discharging demands. While this may be the strongest assumption, our optimization model incorporates a cost to each truck; hence, the solution will output a limited number of trucks. We hold this assumption for two reasons: first, this helps meet the feasibility constraints, easing the solving process. Second, although there may be situations in which the number of trucks has already been decided, we consider the case in which the planner does not know how many electric trucks to purchase. Assuming that we start from an unlimited number of trucks and arriving at a solution will help determine the number of trucks to purchase.

In terms of charging, we will allow multiple charging activities, whether they be charging or discharging, and any charging activity will allow partial recharges or discharges.

Each task will have starting and ending times starting and ending locations, and (if any) charging activities that occur at specific stations. Overnight charging may not be sufficient to perform all the assigned tasks, thus charging may be necessarily included in the schedule. Furthermore, when there is surplus of solar power in comparison to electric demand, the trucks should store energy both to serve future demand and to be ready to travel so that the use of fossil fuel is minimized. The latter, in addition to covering all tasks, is the goal of the EVSP problem.

4 Vehicle Scheduling Problem Formulation

The Vehicle Scheduling Problem is a combinatorial optimization problem in transportation logistics, where a fleet of vehicles must be scheduled to cover a set of trips at minimum operational cost. Each feasible route in the VSP can be represented as a column in a set-partitioning formulation, indicating which trips a vehicle covers.

Column Generation Column generation is a decomposition technique designed to efficiently handle large-scale linear programs with an exponential number of variables, such as those arising from routing problems. It operates by iteratively solving the linear relaxation of the restricted master problem (RMP) and generating new columns with negative reduced cost via a pricing subproblem. While its modern formulation is often attributed to the foundational work of Dantzig and Wolfe [4] and subsequent applications such as Gilmore and Gomory [9], earlier forms of the technique can be traced to the work of Kantorovich and Zalgaller [14], among others. Column generation is inherently a methodology for solving linear programs (LPs); thus, it solves only the LP relaxation of the VSP. To obtain an integer solution, it is typically integrated into a branch-and-price framework or combined with heuristic strategies.

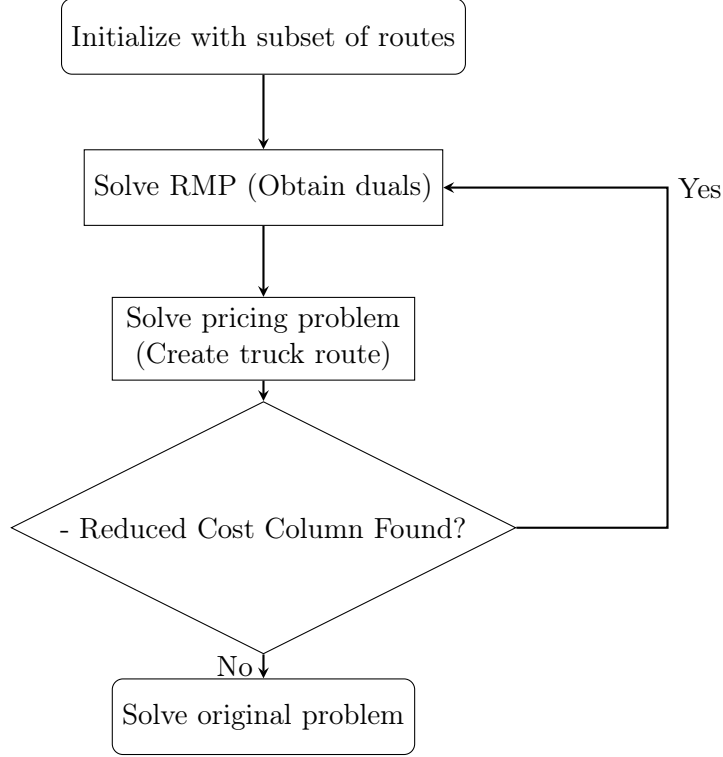


Fig. 1 Flowchart of the Column Generation Algorithm

To clarify the approach, Fig. 1 outlines the iterative column generation algorithm used in our study. We initially construct feasible columns, solve the linear relaxation of the RMP optimally, and then use the resulting dual values to guide the pricing subproblem, identifying improving routes iteratively. This iterative process continues until no more improving columns are found. In our implementation, we adopt a variant that iteratively generates columns and solves the final RMP to integrality using integer programming; that is, we do not use branch and price.

Notation and Parameters We introduce parameters, sets, and decision variables. Let the set of scheduled tasks (or trips) and depot locations be given, with a set of prescribed times by which each task must be completed at its destination. Naturally, the distances between the tasks and the depot where all trucks begin their routes are also assumed to be known; we incorporate this using our choice of a distance function denoted by d .

The objective of the VSP is to find the trip assignment that minimizes the cost from the system perspective. The set of tasks is denoted as T , and each trip $i \in T$ is characterized by its start time st_i , end time et_i , and the corresponding start and end locations sl_i and el_i . The transition between trips is captured by the distance d_{ij} , which represents a composite measure of the time and energy (which are proportional to d_{ij}) required from the end location of trip i to the start location of trip j ; d can be generalized to the distance across stations where the trucks can refuel (charge in the EVSP problem), trips, and the origin depot O , if the vehicle can do that at locations other than the original depot. The fixed cost of deploying a truck is denoted by $\overline{c}_b$. In the classic VSP, trucks are internal combustion engine vehicles, thus the energy cost is associated with the fuel that is needed to cover the trip.

The set of feasible routes is defined as R . In practice, there are exponentially many elements in the set R . Thus, our column generation algorithm starts with a restricted subset $R' \subset R$ selected

for the master problem.

Before presenting the model, we look at an example to see how the solution variables can be visualized. We consider two feasible trip routes: Route r_1 serves trip 1 and trip i , while route r_2 serves trips j and $|T|$. This scenario is illustrated in the matrix equation (1). In the set partitioning problem, the binary decision variable a_r equals 1 if route $r \in R'$ is chosen, and parameter $\delta_{r,i}$ is set to 1 if trip $i \in T$ is covered by route r . In equation (1), we see that the routes are defined in such a way that $\delta_{r_1,1} = 1, \delta_{r_1,i} = 1, \delta_{r_2,j} = 1, \delta_{r_2,|T|} = 1$, while all other δ values are 0 for the two routes. Fig. 2 depicts the two considered routes, where copies of depot O are labeled A and Ω for the pull out and pull in arcs, respectively.

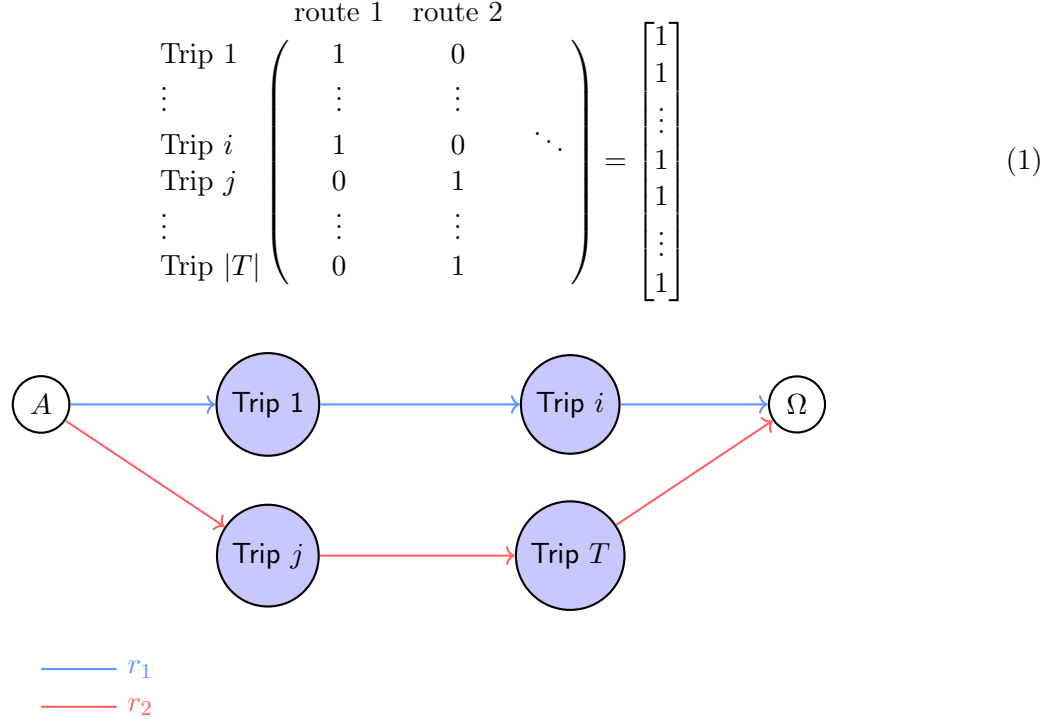


Fig. 2 Routes r_1 and r_2 covering specified trips

Set Partitioning Model The set partitioning problem aims to select the lowest-cost set of routes covering all trips. This is referred to as the *master problem*, namely

$$\text{Minimize } \sum_{r \in R} c_r a_r \quad (2)$$

s.t.

$$\sum_{r \in R} \delta_{r,i} a_r = 1, \quad \forall i \in T \quad (3)$$

$$a_r \in \{0, 1\}, \quad \forall r \in R. \quad (4)$$

Due to constraints (3), we can relax constraint (4) to be $a_r \in \mathbb{Z}_{\geq 0}$. For simplicity, we initially define the cost of the route $c_r = \overline{c}_b$ to a constant cost of utilizing a truck. In that way, we are solving the vehicle scheduling problem by minimizing the number of trucks required to perform every trip.

We begin by solving the linear programming relaxation of the master problem by further relaxing constraint (4) to

$$a_r \geq 0 \quad \forall r \in R \quad (5)$$

Restricted Master Problem The linear relaxation of the RMP will be the same as the linear relaxation of the Master Problem, except on a feasible subset, $R' \subset R$.

We denote with α the dual variables associated with constraints (3).

Pricing Problem To generate a new improving route, namely, a route that could reduce the total cost of the current solution, we introduce variables that define feasible paths. A binary variable x_{ij} represents the feasibility of transitions between trips $i \in T$ and $j \in T$, w_i^A define the vehicle pull-out of the depot to trip $i \in T$, and w_i^Ω defines the pull-in arc from trip $i \in T$ to the depot.

Fig. 2 illustrates feasible route constructions. For example, route r_2 has $w_j^A = 1$, $x_{jT} = 1$, and $w_T^\Omega = 1$, with all other variables equal to 0.

Given the optimal dual solution α^* associated with constraints (3), we solve the pricing subproblem to find improving routes $\bar{r} \in R \setminus R'$.

Note that for a given route $\bar{r}$, $\delta_{\bar{r},i} = w_i^A + \sum_{j \in T} x_{ji}$, that is, a trip $i \in T$ is covered by route $\bar{r}$ if i is either a pull-out from the depot, or it follows another trip $j \in T$.

Thus,

$$c_{\bar{r}} - \sum_{i \in T} \alpha_i^* \delta_{\bar{r},i} = \bar{c}_b - \left[\sum_{i \in T} \alpha_i^* (w_i^A + \sum_{j \in T} x_{ji}) \right] \quad (6)$$

and we solve the mixed integer linear program (MILP)

$$\text{Minimize: } \bar{c}_b - \left[\sum_{i \in T} \alpha_i^* (w_i^A + \sum_{j \in T} x_{ji}) \right] \quad (7)$$

$$\text{s.t.: } \sum_{i \in T} w_i^A = \sum_{i \in T} w_i^\Omega = 1 \quad (8)$$

$$w_i^A + \sum_{j \in T} x_{ji} = w_i^\Omega + \sum_{j \in T} x_{ij}, \quad \forall i \in T \quad (9)$$

$$et_i + d_{ij} \leq st_j + M(1 - x_{ij}), \quad \forall i, j \in T \quad (10)$$

$$x_{ij}, w_i^A, w_i^\Omega \in \{0, 1\}, \quad \forall i, j \in T. \quad (11)$$

Constraint (8) ensures that each feasible route begins by pulling out from the depot and ends by pulling in to the depot. Constraints (9) enforces flow conservation, ensuring that trips are linked correctly within a route. If a trip is entered from a depot or another trip, it must also be exited (either to another trip or back to the depot). Constraints (10) ensures temporal feasibility by requiring that if trip i is followed by trip j , the earliest end time of i plus travel time to j does not exceed the latest start time of j . Finally, the integrality constraints (11) ensures that decision variables indicating trip transitions and route endpoints are binary, yielding feasible routes.

Our method does not use branch-and-price; instead, once no more routes with negative reduced cost are found, we reintroduce the integrality constraints and solve the final set partitioning problem over the RMP using only the generated columns.

5 The Microgrid EVSP-V2G problem formulation

We now extend the VSP to include partial charging and discharging in a closed microgrid. Building upon the previously detailed assumptions and framework, we extend the classic VSP to the Electric Vehicle Scheduling Problem with Vehicle-to-Grid capability (EVSP-V2G).

In the EVSP-V2G formulation, not all feasible vehicle routes necessarily include trip coverage. Some feasible routes may consist solely of charging and discharging actions, designed to shift energy across time and alleviate demand peaks. We refer to these as *battery schedules*. Their inclusion in the model serves both operational and planning purposes: they enable additional flexibility in utilizing renewable energy and can represent stationary battery deployments. This distinction will become important later when we differentiate between truck routes and battery schedules.

To model charging and discharging decisions, we introduce a discrete time horizon divided into $\bar{t}$ time blocks, indexed by $t \in [\bar{t}]$. Let S denote the set of all charging stations. For each route $r \in R$, charging station $h \in S$, and time period $t \in [\bar{t}]$, we define binary parameters to indicate charging and discharging activity.

$$\psi_{r,h,t}^{+free}, \psi_{r,h,t}^+, \psi_{r,h,t}^-, \psi_{r,h,t}^{-free} \in \{0, 1\},$$

where each value is determined by the structure of the route r as follows:

- $\psi_{r,h,t}^{+free} = 1$ if route r includes free (excess solar) charging at station h and time t ; 0 otherwise.
- $\psi_{r,h,t}^+ = 1$ if route r includes paid charging (from electricity generated by fossil fuels) at station h and time t ; 0 otherwise.
- $\psi_{r,h,t}^- = 1$ if route r includes discharging to the grid (V2G) at station h and time t ; 0 otherwise.
- $\psi_{r,h,t}^{-free} = 1$ if route r includes discharging to another vehicle (V2V) at station h and time t ; 0 otherwise.

The net power demand at time t is defined as

$$\Delta_t = D_t - \Sigma_t, \quad \forall t \in [\bar{t}],$$

with D_t and Σ_t denoting known electricity demand and solar generation, respectively. We denote the positive and negative parts of net demand

$$\Delta_t^+ := \max\{0, \Delta_t\}, \quad \Delta_t^- := \max\{0, -\Delta_t\}, \quad \forall t \in [\bar{t}].$$

We start with the most natural formulation by minimizing total electricity generation and vehicle usage costs. Letting c_g denote the cost of per unit electricity generation via fossil fuel, and defining

ρ as the rate of charging or discharging, we have

$$\text{Minimize } c_g \sum_{t \in [\bar{t}]} gen_t + \bar{c}_b \sum_{r \in R} a_r \quad (12)$$

s.t.

$$gen_t = \Delta_t^+ + \rho \sum_{r,h} a_r (\psi_{r,h,t}^+ - \psi_{r,h,t}^-), \quad \forall t \in [\bar{t}] \quad (13)$$

$$gen_t \geq 0, \quad \forall t \in [\bar{t}] \quad (14)$$

$$\sum_r a_r \delta_{r,i} = 1, \quad \forall i \in T \quad (15)$$

$$\rho \sum_{r,h} a_r \psi_{r,h,t}^- \leq \Delta_t^+, \quad \forall t \in [\bar{t}] \quad (16)$$

$$\rho \sum_{r,h} a_r (\psi_{r,h,t}^{+free} - \psi_{r,h,t}^{-free}) \leq \Delta_t^-, \quad \forall t \in [\bar{t}] \quad (17)$$

$$a_r \in \{0, 1\}, \quad \forall r \in R. \quad (18)$$

Constraints (13) define the net fossil generation gen_t at each time period t , while non-negativity of electricity generation is enforced by constraints (14). The trip coverage constraints (15) are identical to the constraints (3) in the VSP. Constraints (16) ensure that discharging into the grid does not exceed the net positive demand Δ_t^+ , while constraints (17) limit opportunistic free charging during periods where more solar energy is available than the base demand. Constraints (15) allow us to relax binary constraints (18) to be $a_r \in \mathbb{Z}_{\geq 0}$.

Observe that when considering only the generation cost in the objective and the fossil generation balance constraints (13)–(14), the problem reduces to a simplified version of the power system's economic dispatch problem, without generator capacity constraints. The inclusion of route costs and constraints (15)–(18) integrates the vehicle scheduling problem, coupling transportation and grid operation decisions.

Furthermore, for each time $t \in [\bar{t}]$, constraint (14) enforcing non-negativity of generation is redundant. This follows because gen_t is defined in constraint (13) as the sum of a nonnegative term Δ_t^+ and the net grid charging minus discharging, which by constraint (16), in addition to binary constraints (18), will force generation to be non-negative.

Accordingly, the problem can be simplified to

$$\text{Minimize } c_g \sum_{t \in [\bar{t}]} \left(\Delta_t^+ + \rho \sum_{r,h} a_r (\psi_{r,h,t}^+ - \psi_{r,h,t}^-) \right) + \bar{c}_b \sum_{r \in R} a_r \quad (19)$$

s.t. (15) – (18).

Implementation To avoid situations where vehicles engage in circular charging and discharging among themselves without contributing meaningful energy exchange, we introduce a small penalty factor $\eta > 0$. This can be interpreted as the energy loss incurred in such round-trip transfers. If $c_{h,t}$ is the cost of generating electricity at station h at time t , then we can define the cost of the route c_r as

$$c_r = \bar{c}_b + \rho \sum_{h,t} c_{h,t} \left((1 + \eta) \psi_{r,h,t}^+ - \psi_{r,h,t}^- \right).$$

and removing the constant generation term, we arrive at the formulation

$$\text{Minimize } \sum_r c_r a_r \quad (20)$$

s.t. (15) – (18).

Furthermore, depending on the given data values regarding solar power and electricity demand, some routes may consist only of charging and discharging, without any trips being performed. The reasons to differentiate between routes that only perform charging and discharging from routes that perform tasks, that is, cover trips, as well as charge and discharge are first that for the former we can perform those tasks with a simple battery, which can cost less than a battery powered truck. And second, on the computational side, these battery schedules will have a much simpler pricing problem since they are only be constrained by charging and discharging of the battery. Thus, for the EVSP-V2G, we use $a_r = 1$ if truck route $r \in R_{\text{truck}}$ is selected, and $b_r = 1$ if $r \in R_{\text{battery}}$ is selected. The associated costs are also defined accordingly: $c_r^a = \overline{c}_b + \sum_{h,t} c_{h,t} \left((1 + \eta) \psi_{r,h,t}^+ - \psi_{r,h,t}^- \right)$, as before, while $c_r^b = \overline{c}_{\text{batt}}$, where $\overline{c}_{\text{batt}}$ is the cost of utilizing a battery. Thus, the resulting RMP is

$$\text{Minimize } \sum_{r \in R_t} c_r^a a_r + \sum_{r \in R_b} c_r^b b_r \quad (21)$$

s.t.

$$\sum_{r \in R_t} \delta_{r,i} a_r = 1, \quad \forall i \in T \quad (22)$$

$$\sum_{r \in R_t} \sum_{h \in S} \psi_{r,h,t}^- a_r + \sum_{r \in R_b} \sum_{h \in S} \psi_{r,h,t}^- b_r \leq \Delta_t^+, \quad \forall t \in [\bar{t}] \quad (23)$$

$$\sum_{r \in R_t} \sum_{h \in S} (\psi_{r,h,t}^{+free} - \psi_{r,h,t}^{-free}) a_r + \sum_{r \in R_b} \sum_{h \in S} (\psi_{r,h,t}^{+free} - \psi_{r,h,t}^{-free}) b_r \leq \Delta_t^-, \quad (24)$$

$$\forall t \in [\bar{t}]$$

$$a_r \in \mathbb{Z}^+ \quad \forall r \in R_t \quad (25)$$

$$b_r \in \mathbb{Z}^+ \quad \forall r \in R_b. \quad (26)$$

Note that the binary variables a_r were relaxed in constraints (25), since constraints (22) ensure that only one truck covers each trip. Variables b_r could naturally be assumed to be integer, as the net demand profile may require deploying multiple batteries to absorb excess solar energy at certain times and discharge it during peak demand periods.

Finally, by relaxing constraints (25) and (26), we solve the pricing problem within the column generation framework applied to the linear relaxation of the RMP. For the complete MILP formulation of the pricing problem, we refer the interested reader to the [Appendix](#).

6 Computational Experiments

Our computational experiments are based on publicly available datasets, scaled appropriately to reflect realistic operational scenarios for isolated microgrids or remote military installations. Demand and solar generation profiles are adapted from Almeida et al. [1], scaled proportionally to match our considered total daily electricity demand.

Based on the feedback of experts in the field, we consider a microgrid serving a small, remote military base, which might be located on an isolated island or in a desert environment. Such isolated

locations typically face significant logistical challenges, including costly and potentially vulnerable supply lines for fuel transportation. To mitigate both economic costs and strategic vulnerabilities, it is natural to explore solutions that maximize reliance on locally produced renewable energy, particularly solar power. Within this context, our objective is to decrease the fuel consumption on the base and increase its reliance on solar energy to support both the energy needs for base operations and vehicle fleet management. Surplus solar energy generated during midday hours can thus be stored in EV batteries for later use.

We consider vehicle trips in which trucks depart from a central depot, complete assigned tasks between specified locations, charge or discharge energy as needed, and return to the depot. These trips can represent scheduled patrols, supply missions, or similar logistic activities. Each trip is approximately two hours in duration and characterized by significant fuel or energy usage, representative of typical heavy-duty military vehicle operations. Factors such as steep terrain, off-road conditions, and heavy payloads further reduce energy efficiency, reflecting realistic operational demands.

For numerical tests, we generated scenarios varying the number of possible starting and ending locations of trips, charging stations, charging stop limits, and number of vehicle trips. Specifically, experiments were conducted with 2 to 4 location points and with 20 to 120 trips and different charging scenarios. Each scenario is defined by generated trips within fixed intervals, where start and end locations were selected from a defined set of nodes. The trips were scheduled within discrete hourly time blocks spanning from 00:00 to 23:00 hours. The energy required per trip was set at 200 kWh.

The depot and charging stations were located at the origin point (0,0) to simplify the network layout, focusing the analysis on the optimization of energy consumption and route efficiency. Distances between nodes were calculated using Manhattan distance, with vehicles traveling at a consistent speed of 1 distance unit per time block.

Charging cost data was uniform across all time blocks, and operational costs were composed of fixed costs for utilizing a truck or a battery, with trucks costing more than a battery. This is realistic in the military context we consider.

The solution was reached by using a column generation approach, iteratively solving the master problem and pricing subproblem until no more improving routes were found. From there, i.e., after the solution of the initial continuous relaxation, the final solutions were obtained using mixed integer programming optimization, with a time limit of 4 hours per scenario.

Trip Parameters and Operational Scenarios To systematically examine the performance of EVs versus ICE vehicles under the described operational conditions, we adopt a set of standardized trip parameters summarized in Table 1. Each logistic mission consumes approximately 15–25 gallons of gasoline (ICE vehicles) or equivalently 150–250 kWh (EVs), depending on intensity. Operational intensities are classified into three distinct scenarios: light (15 gal/150 kWh), medium (20 gal/200 kWh), and heavy (25 gal/250 kWh), denoted by the parameter $\epsilon \in \{1.5, 2.0, 2.5\}$, respectively. In all scenarios with varying intensities, the trip duration is held constant at 2 hours.

Energy comparisons are standardized using an energy conversion factor of 33 kWh per gallon of gasoline. Electricity use is consistently converted into equivalent gallons of fuel, under the assumption that any unmet electricity demand in an isolated microgrid without external support must be provided by fossil-fuel generation.

Demand and Generation Profiles The hourly base electricity demand profile is derived from real-world data for a military installation located on San Nicolas Island [19], which reports peak

Parameter	Gas Vehicle	Electric Vehicle
Trip duration	2 h	2 h
Battery capacity	–	700 kWh
Light-load energy use per trip	15 gal	150 kWh
Medium-load energy use per trip	20 gal	200 kWh
Heavy-load energy use per trip	25 gal	250 kWh

Table 1: Key parameters used in trip settings

loads of approximately 975 kW and minimum base loads around 500 kW throughout the day. We use this information to scale a load profile such that the resulting total daily demand integrates to approximately 20 MWh.

Rather than explicitly modeling this specific location, our scenario represents typical isolated military bases, ensuring that renewable integration and vehicle-to-grid (V2G) operations are analyzed in a realistic context.

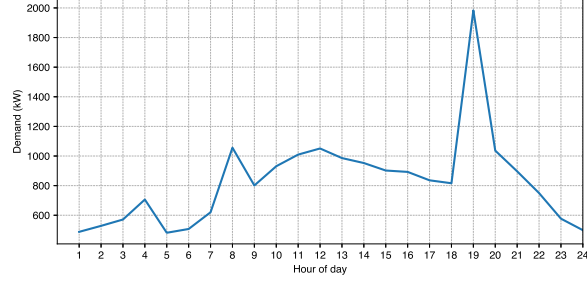
As for solar energy generation, we take a real dataset and scale it accordingly. Specifically, we assume a solar PV installation with a rated capacity of 2.8 MW, which corresponds to a daily generation total of about 14 MWh, using a yield of 5 kWh/day per kW of capacity. This system size would require approximately 14–28 acres of land, which is reasonable for a medium-sized remote base. In our context, we will only be interested in cases where the solar generation profile is strictly greater than the demand profile; otherwise, there is not enough excess solar energy to charge the EVs and the EVSP gets reduced to an uninteresting problem. It is only when solar energy exceeds the demand at some point that the EVSP-V2G is interesting enough to solve.

Figure 3 illustrates temporal distributions of demand, solar output, and their net balance, highlighting periods of surplus and deficit throughout the day.

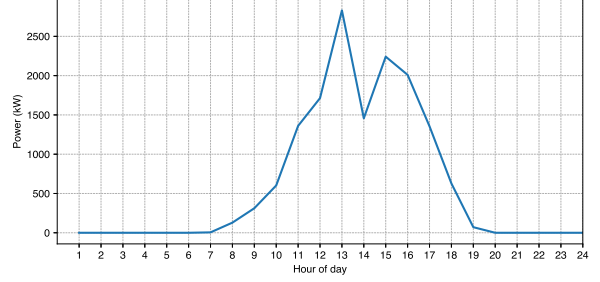
Fig. 3(a) presents the base’s hourly demand curve, which integrates to roughly 20 MWh over 24 hours. Fig. 3(b) shows the solar PV output profile, totaling about 14 MWh. In Fig. 3(c), we overlay these two series to highlight that solar meets only a portion of the load. Finally, Fig. 3(d) plots the difference (solar minus demand), revealing a midday surplus of approximately 500 kWh and deficits during morning and evening. Without sufficient battery storage to capture the midday excess, the microgrid cannot fully utilize the 14 MWh of solar energy against the 20 MWh demand.

Comparative Cost-Benefit Analysis of V2G Operations A practical consideration is whether employing EVs yields tangible benefits, independently of environmental or strategic advantages. To perform a rigorous cost-benefit analysis, we compare the performance of three operational paradigms, namely VSP, EVSP-Solar, and EVSP-V2G. In particular, reducing the need for refueling the base not only brings economic savings but also strategically mitigates the risks associated with reliance on vulnerable fuel supply chains.

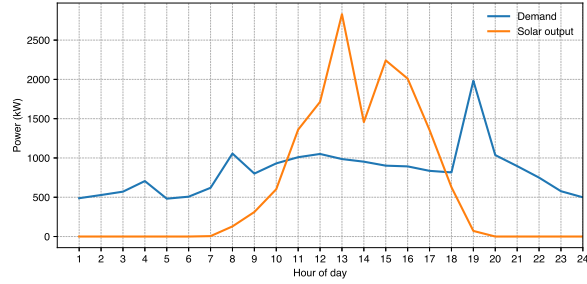
In the baseline VSP scenario, vehicles powered by ICE cannot charge or discharge electricity; hence, they strictly minimize operating costs subject only to trip coverage and integer vehicle constraints. By contrast, in the EVSP-Solar scenario, EVs can recharge freely from surplus solar energy but must otherwise rely on fossil-fuel-generated electricity, incurring corresponding costs. Technically, this is enforced through constraints (24), allowing charging only during periods of solar surplus. Finally, the EVSP-V2G scenario extends this flexibility, enabling vehicles to not only charge from solar surpluses or other vehicles’ discharges but also discharge energy back into the microgrid



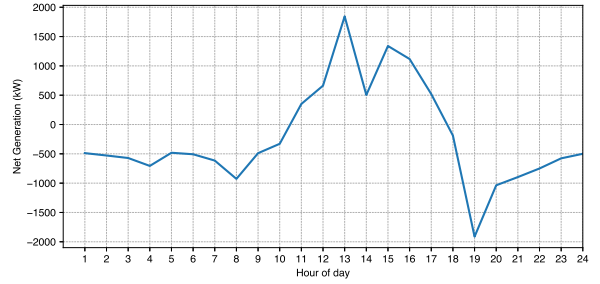
a Hourly electricity demand (kW)



b Total solar output (kW)



c Demand vs. solar output over 24 h (kW)



d Net generation demand-solar (kW)

Fig. 3 Temporal profiles over one day: (a) hourly demand, (b) solar output, (c) demand vs. solar, (d) net generation

when beneficial. This advanced capability is explicitly represented by constraints (23).

From Fig. 4, we observe that both EVSP-Solar and EVSP-V2G consistently outperform the baseline VSP scenario in terms of fuel savings. This result aligns intuitively with the ability of EVs to harness renewable energy.

Conversely, a different picture emerges when examining fleet size. Fig. 4 reveals that both EV-based scenarios typically require a larger fleet compared to the ICE-based VSP solution. This is primarily due to limited EV range and charging constraints, requiring additional vehicles to ensure uninterrupted operations. Under heavy-duty conditions, an interesting result emerges, and we observe that EVSP-V2G sometimes satisfies operational demands using fewer vehicles than EVSP-Solar. This outcome reflects the enhanced scheduling flexibility introduced by V2G capability, which allows vehicles with surplus battery capacity to discharge into the microgrid and indirectly support other vehicles' charging needs. This additional flexibility can offset the higher energy requirements associated with heavy-duty tasks and reduce the total number of vehicles needed.

These results highlight that the adoption of EV technology, particularly with V2G capabilities, can offer substantial economic and operational advantages in isolated microgrid scenarios, provided that fleet sizing and scheduling complexities are adequately managed.

Impact of Trip Scheduling Given midday solar surpluses observed in Fig. 3(d), strategic trip scheduling by avoiding peak solar hours could further enhance fuel savings. Table 2 compares uniformly distributed trips versus scheduled breaks around midday peaks, indicating additional fuel savings through optimized scheduling.

Keeping the total number of trips constant, we compare two scheduling scenarios in Table 2: *uniform*, where trips are evenly distributed during standard working hours (9:00–18:00), and *scheduled*

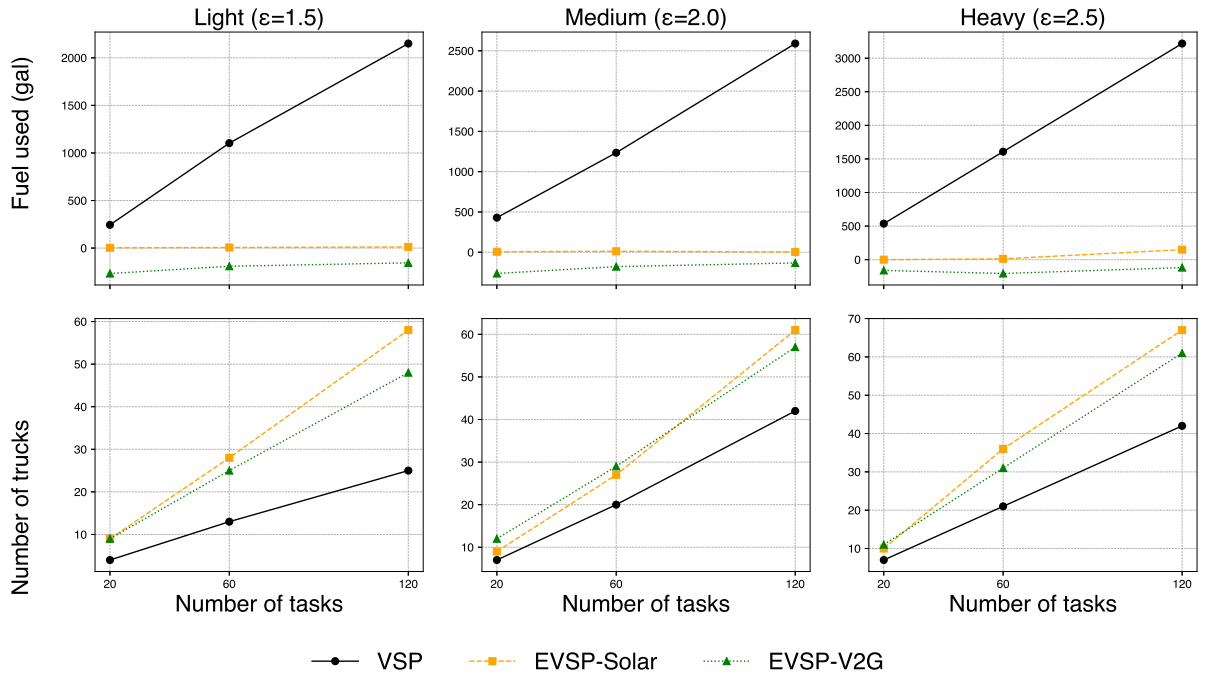


Fig. 4 Comparison of fuel usage (top row) and fleet size (bottom row) across three operational intensities left to right: light ($\epsilon = 1.5$), medium ($\epsilon = 2.0$), heavy ($\epsilon = 2.5$). Markers denote VSP (solid circle), EVSP-Solar (dashed square), and EVSP-V2G (dotted triangle)

breaks, where trips are concentrated before 9:00 and after 18:00, specifically within the windows 4:00–9:00 and 18:00–23:00. Both scenarios have identical total scheduling windows of 10 hours, but the *scheduled-breaks* approach aligns better with solar availability and yields greater fuel savings. Such scheduling adjustments may be particularly practical for military contexts, where operational flexibility and extended working hours are often feasible.

Table 2: Mean daily fuel usage (gallons) and mean fleet size for $\epsilon = 2.5$ under two scheduling scenarios: scheduled-breaks (trips scheduled within 4:00–9:00 and 18:00–23:00) versus uniform scheduling (trips uniformly scheduled during 9:00–18:00). Negative fuel indicates net export to the base load

Trips	Fuel (gal)		Trucks	
	Breaks	Uniform	Breaks	Uniform
20	-158.59	-128.28	12.33	12.00
60	-60.61	-29.29	30.67	25.67
120	-57.58	-6.06	62.00	55.00

Impact of Increasing Solar Energy Next, we examine the effect of increasing the available solar energy on fleet operations. Figure 5 shows how scaling up the available daily solar energy affects the required fleet size and net fuel consumption under medium- and heavy-duty demand scenarios.

Figure 5 shows that as the amount of solar energy grows, the fleet realizes larger fuel savings; in the V2G-enabled cases the net fuel used becomes negative, reflecting that surplus stored energy is being returned to satisfy base demand and reduce conventional fuel burning. When fuel prices are sufficiently high, deploying additional vehicles to absorb, store, and redistribute solar energy can be economically justified because it adds flexibility that improves utilization of renewable generation. However, there is a point of diminishing returns: beyond a certain level of daily solar availability, further increases do not yield additional fuel savings. This saturation occurs because the daily demand profile and the one-day optimization horizon limit how much stored solar energy can be discharged to displace fuel. Once the base load is met, extra solar energy cannot further reduce fuel use unless demand increases along with the available solar.

Solution Example Figure 6 shows the detailed timeline of charging and discharging actions across all selected routes in the EVSP-V2G model. Each horizontal band corresponds to one route, with paid charging represented by solid fill, free (solar) charging by diagonal hatch, and discharging by cross hatch. This visualization confirms that the model leverages free solar energy whenever available and schedules discharges primarily during high-generation periods.

Computation Under Different Instance Sizes Tables 3 and 4 summarize the computational performance of the EVSP-V2G model under two different energy availability settings: $\epsilon = 2$ and $\epsilon = 1.5$, respectively. In both cases, we observe a predictable increase in runtime, number of vehicles, and columns generated as the number of locations and associated tasks increases.

For the more relaxed energy constraint $\epsilon = 1.5$, the solver takes significantly longer for the same number of tasks compared to the more energy-intensive setting of $\epsilon = 2$. This can be explained by the fact that lower ϵ allows each EV to use less charge per task, which in turn enables longer sequences of tasks per vehicle. Empirically, we observe that trucks perform approximately four tasks per route when $\epsilon = 1.5$, compared to around three when $\epsilon = 2$. Since each task spans two

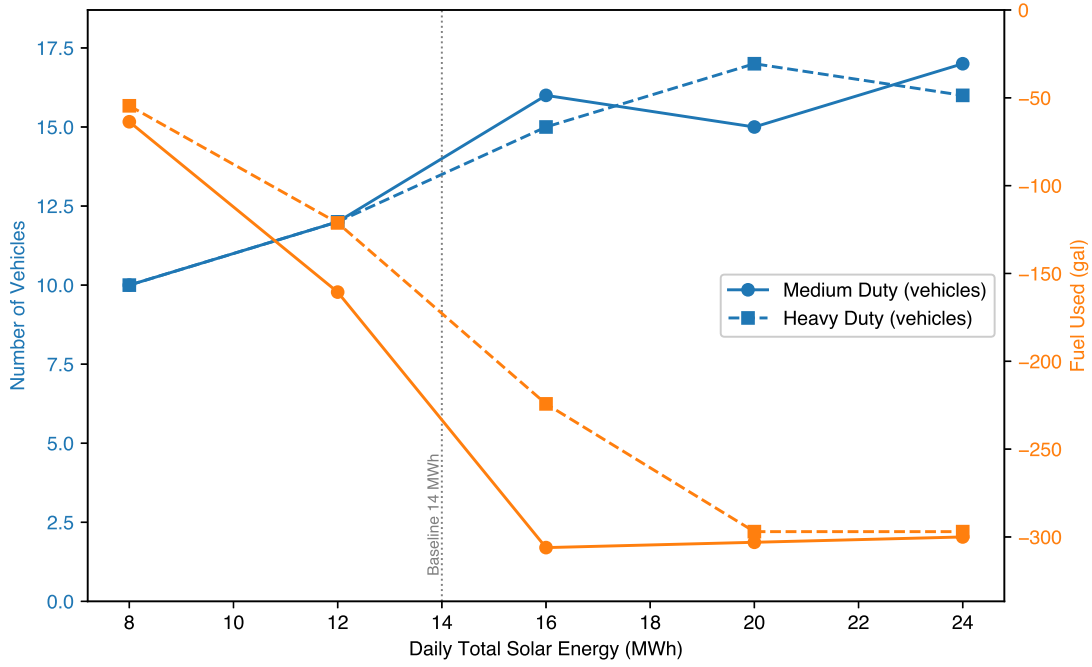


Fig. 5 Impact of available daily solar energy (MWh) on fleet size and net fuel consumption for medium-duty ($\epsilon = 2.0$) and heavy-duty ($\epsilon = 2.5$) vehicle scenarios. The x-axis reports the total daily solar energy made available by scaling the base generation profile; the vertical dotted line indicates the default daily solar energy of 14 MWh used in all other experiments. Left vertical axis shows the number of vehicles required; right vertical axis shows the net fuel used in gallons. Negative fuel values correspond to net discharge from EVs with V2G back to the base load, offsetting what would otherwise be supplied by fossil fuel generation

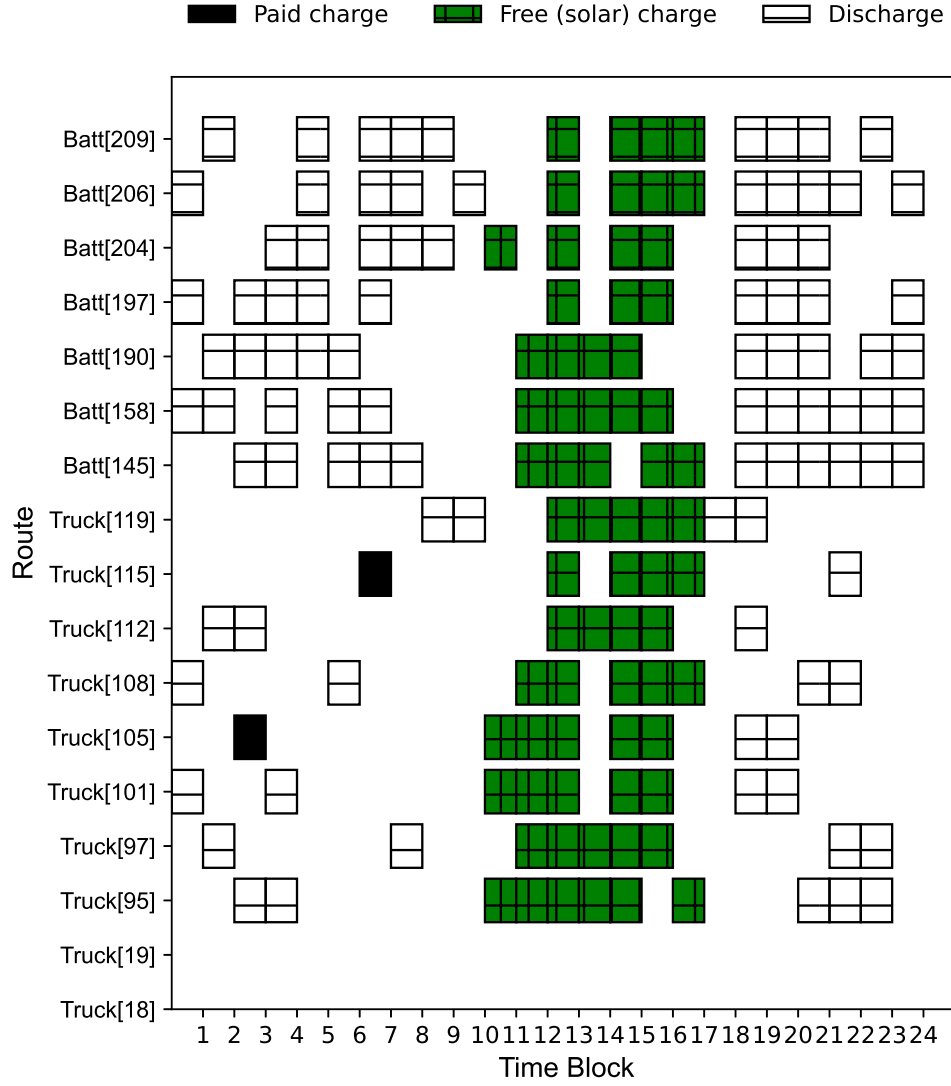


Fig. 6 Charging and discharging timeline for each selected route over 24 time blocks. Solid black bars indicate paid charging, "+" hatch indicates free (solar) charging, and "-" hatch indicates discharge events

hours, this task density per vehicle is consistent with a 10 hour scheduling horizon and confirms the model’s ability to efficiently exploit additional flexibility when feasible.

Despite the additional complexity, the optimality gap remains below 1% in all instances, and exact solutions are reached in several cases. The pricing subproblem, which generates the feasible routes, dominates the total runtime in larger instances, where it regularly exceeds 95% of total solve time.

Finally, the difference in the number of locations and tasks reported between Tables 3 and 4 reflects computational tractability limitations under the more flexible setting. A full breakdown of solver diagnostics, including LP and MIP statistics, is provided in Tables 5 and 6 in the [Appendix](#) for completeness.

Table 3: Sparse-trip morning runs for the EVSP–V2G model with up to 10 locations ($\epsilon = 2$). Each row reports: the number of locations; the total number of trip tasks; the overall solution time (seconds); the fraction of computational time spent in the master-problem, pricing-subproblem, and MIP-solve; the total number of columns (EV and battery scheduling routes) generated; the number of vehicles deployed; and the final optimality gap.

Loc.	Tasks	Time (s)	Master (%)	Pricing (%)	MIP (%)	Cols	Vehicles	Gap (%)
2	10	26	1.96	95.74	2.30	16333	4	0.00%
3	30	53	0.68	97.60	1.72	16152	11	1.00%
4	60	140	0.51	98.32	1.17	20694	21	0.49%
5	100	537	0.27	95.44	4.29	21361	34	0.66%
6	150	817	0.14	99.44	0.41	31273	50	0.48%
7	210	2688	0.11	99.78	0.11	45957	70	0.68%
8	280	4378	0.04	99.93	0.04	28965	94	0.90%
9	360	14224	0.05	99.93	0.03	55169	121	0.93%
10	450	29722	0.03	99.95	0.02	45906	151	0.71%

Table 4: Sparse-trip morning runs for the EVSP–V2G model with up to 8 locations ($\epsilon = 1.5$). See Table 3 for column definitions.

Loc.	Tasks	Time (s)	Master (%)	Pricing (%)	MIP (%)	Cols	Vehicles	Gap (%)
2	10	40	1.19	91.46	7.35	17606	3	0.00%
3	30	611	3.93	88.86	7.21	52490	9	0.88%
4	60	950	1.17	98.78	0.05	39953	15	0.00%
5	100	3800	0.56	99.42	0.03	39545	25	0.00%
6	150	5808	0.22	98.20	1.58	34862	38	0.86%
7	210	13841	0.24	99.74	0.02	68020	53	0.94%
8	280	21433	0.14	94.70	5.16	49559	71	0.94%

7 Conclusion

In this paper, we presented an extension of the VSP incorporating V2G capabilities, specifically designed for microgrids with substantial solar energy integration. Our model jointly addresses logistical scheduling and energy constraints, allowing electric vehicles to meet transportation demands while

optimizing the use of renewable solar energy. By strategically managing charging and discharging schedules, the EVSP-V2G framework reduces reliance on fuel-based generation and enhances the resilience and flexibility of isolated microgrids, delivering strategic, economic, and environmental benefits.

We employed a column generation approach to solve large-scale instances efficiently and demonstrated performance improvements in fuel savings and solar energy utilization.

Future research directions include extending the model to civilian grid settings, where vehicle scheduling and V2G operations could affect electricity market prices and system-wide energy planning. Incorporating stochastic elements such as uncertain weather and demand profiles would further improve robustness. Finally, a promising direction involves integrating detailed grid constraints, such as voltage limits, AC power flow equations, thermal loading, and dispatchable generation limits, to better align operational scheduling with power systems engineering requirements.

Statements

Author contributions

Nathan Cho developed the optimization model for electric vehicle scheduling with V2G integration, implemented the codebase, designed the experimental settings and scenarios, and conducted all computational experiments. Andrea Lodi contributed to the modeling of the EVSP and provided key algorithmic insights, including the use of fast pricing columns and battery routes to improve computational performance and prevent redundant route generation. Anna Scaglione initiated the research direction by proposing the real-world motivation involving V2G in electrical grids, and provided guidance on integrating grid-side constraints and renewable energy interactions into the model.

Data availability

The code and input data needed to reproduce the results of this paper are publicly available at <https://github.com/ndandnd/evspv2g>. All computational experiments were implemented in Python 3.12.2 and solved using Gurobi Optimizer 12.0.1 (academic license) for all mixed-integer and linear programs.

Acknowledgments

The authors gratefully acknowledge Prof. Tong Wu, Department of Electrical and Computer Engineering, University of Central Florida, for guidance and for preparing the data used in our model.

8 Appendix

8.1 Pricing Problems

8.1.1 MILP Formulation of the EVSP-V2G Pricing Problem

Before we present our MILP for the pricing problem, we note that for the classical EVSP, Desaulniers et al. [6] demonstrate that dynamic programming with labeling algorithms can be used to efficiently solve the pricing problem. Routes are extended by favoring those that arrive at trips earlier and with a higher state of charge. However, this approach assumes static electricity pricing and does not account for external renewable energy availability. In our EVSP-V2G setting, this assumption

no longer holds. When excess solar energy is available, higher state-of-charge routes may be less desirable, as they miss opportunities for free charging. Conversely, during periods of higher electrical demand, higher state-of-charge routes are preferred for potential discharging. Due to this temporal dependency on solar conditions and the need to model complex charging decisions, we find that the traditional labeling approach does not translate well. Instead, we formulate the pricing problem as a MILP, which allows greater modeling flexibility and alignment with grid-aware scheduling objectives. Our MILP model was able to handle test cases of up to 450 trips, but future work could focus on speeding up the computation for larger datasets.

To solve the master problem in Section 5, we generate new battery-only routes that may improve the objective. Specifically, we solve a pricing subproblem that identifies a feasible battery route that improves our solution using the current dual solution from the linear relaxation of the RMP.

We start from the RMP objective (21), which includes both truck and battery-only routes. Let $\alpha_i^*, \beta_t^*, \gamma_t^*$ be the dual variables corresponding to constraints (22), (24), and (23), respectively. These duals are used to compute the reduced cost of a new route $\bar{r} \in R \setminus R'$, where R' is the set of current battery routes.

When creating a new feasible route in the pricing problem, we introduce indicators that carry the same meaning as the indicators ψ in Section 5, but denote that the route is fixed. For instance, $\psi_{\bar{r}ht}^+ = \chi_{ht}^+$ for a fixed route $\bar{r} \in R$ that is being generated by the pricing problem.

We also introduce new variables and sets associated with the EVSP-V2G setting, partitioning the set of all charging stations S into $S_l \subseteq S$, the subset of stations available for the l^{th} charging or discharging activity, where $l = 1, \dots, L$, and L is the maximum number of allowed charging activities in a route.

We extend the VSP pricing model (7), introducing additional routing binary variables include y_{ih} , which indicates whether the vehicle transitions from trip $i \in T$ to station $h \in S$, and z_{hi} , which captures the transition from station h back to trip i . For temporal modeling, cst_h and cet_h denote the start and end times of the charging or discharging activity at station $h \in S$, respectively.

The battery dynamics are captured using the maximum state of charge G , and the variable g_i represents the state of charge upon arrival at the start of trip i , while ϵ_i denotes the energy required to complete trip i . The net amount of energy charged or discharged at station h is denoted by v_h . These additions allow us to account for energy flows and operational feasibility in the pricing problem formulation for the EVSP-V2G, given by objective (27) and constraints (28)-(57).

$$\begin{aligned}
\text{Minimize } c_{\bar{r}} - & \left[\sum_{i \in T} \alpha_i^* \delta_{\bar{r}i} + \sum_{t \in [\bar{t}]} \sum_{h \in S} \left(\beta_t^* (\psi_{\bar{r}ht}^{+free} - \psi_{\bar{r}ht}^{-free}) + \gamma_t^* \psi_{\bar{r}ht}^- \right) \right] = \\
& \bar{c}_b + \sum_{h \in S} \sum_{t \in [\bar{t}]} c_{h,t} (\chi_{h,t}^+ - \chi_{h,t}^-) - \\
& \left[\sum_{i \in T} \alpha_i^* (w_i^A + \sum_{h \in S} z_{hi} + \sum_{j \in T} x_{ji}) + \sum_{t \in [\bar{t}]} \sum_{h \in S} \left(\beta_t^* (\chi_{ht}^{+free} - \chi_{ht}^{-free}) + \gamma_t^* \chi_{ht}^- \right) \right]
\end{aligned} \tag{27}$$

s.t.

$$\sum_{h \in S_1} w_h^A + \sum_{i \in T} w_i^A = \sum_{h \in \bigcup_{l=1}^L S_l} w_h^\Omega + \sum_{i \in T} w_i^\Omega = 1 \quad (28)$$

$$w_h^A = 0 \quad \forall h \in S_l, l > 1 \quad (29)$$

$$w_i^A + \sum_{j \in T} x_{ji} + \sum_{h_l \in S_l} z_{h_l i} = w_i^\Omega + \sum_{j \in T} x_{ij} + \sum_{h_{l+1} \in S_{l+1}} y_{ih_{l+1}} \quad \forall i \in T, \forall l = 1, \dots, L-1 \quad (30)$$

$$w_h^A + \sum_{i \in T} y_{ih} = w_h^\Omega + \sum_{i \in T} z_{hi} \quad \forall h \in S \quad (31)$$

$$\sum_{h \in S_l} (w_h^A + \sum_{i \in T} y_{ih}) \leq 1 \quad \forall l = 1, \dots, L \quad (32)$$

$$\bar{l} = \sum_{h \in S} (w_h^A + \sum_{i \in T} y_{ih}) = \sum_{h \in S} (w_h^\Omega + \sum_{i \in T} z_{hi}) \leq L \quad (33)$$

$$et_i + \frac{d_{ij}}{\text{speed}} \leq st_j + M(1 - x_{ij}) \quad \forall i, j \in T \quad (34)$$

$$cst_h = w_h^A \frac{d_{Oh}}{\text{speed}} + \sum_{i \in T} y_{ih} (et_i + \frac{d_{ih}}{\text{speed}}) \quad \forall h \in S \quad (35)$$

$$cet_h = w_h^\Omega (\bar{t} - \frac{d_{Oh}}{\text{speed}}) + \sum_{i \in T} z_{hi} (st_i - \frac{d_{hi}}{\text{speed}}) \quad \forall h \in S \quad (36)$$

$$0 \leq cst_h \leq cet_h \leq \bar{t} \quad \forall h \in S \quad (37)$$

$$\chi_{h,t}^+ + \chi_{h,t}^- + \chi_{h,t}^{+free} + \chi_{h,t}^{-free} + \chi_{h,t}^0 = C_{h,t} \quad \forall h \in S, \forall t = 1, \dots, \bar{t} \quad (38)$$

$$t \leq cet_h + M(1 - C_{h,t}) \quad \forall h \in S, \forall t \in [\bar{t}] \quad (39)$$

$$t \geq cst_h - M(1 - C_{h,t}) \quad \forall h \in S, \forall t \in [\bar{t}] \quad (40)$$

$$v_h = \sum_{t=0}^{\bar{t}} (\chi_{h,t}^{+free} + \chi_{h,t}^+ - \chi_{h,t}^- - \chi_{h,t}^{-free}) \quad \forall h \in S \quad (41)$$

$$0 \leq g_i \leq G \quad \forall i \in T \cup S \quad (42)$$

$$0 \leq g_h + v_h \leq G \quad \forall h \in S \quad (43)$$

$$g_O = G \quad (44)$$

$$g_j \leq g_i - \epsilon_i - d_{ij} + M(1 - x_{ij}) \quad \forall i, j \in T \quad (45)$$

$$g_j \geq g_i - \epsilon_i - d_{ij} - M(1 - x_{ij}) \quad \forall i, j \in T \quad (46)$$

$$g_h \leq g_i - \epsilon_i - d_{ih} + M(1 - y_{ih}) \quad \forall i \in T, \forall h \in S \quad (47)$$

$$g_h \geq g_i - \epsilon_i - d_{ih} - M(1 - y_{ih}) \quad \forall i \in T, \forall h \in S \quad (48)$$

$$g_j \leq g_h + v_h - d_{hj} + M(1 - z_{hj}) \quad \forall h \in S, j \in T \quad (49)$$

$$g_j \geq g_h + v_h - d_{hj} - M(1 - z_{hj}) \quad \forall h \in S, j \in T \quad (50)$$

$$g_i \leq g_O - d_{Oi} + M(1 - w_i^A) \quad \forall i \in T \quad (51)$$

$$g_i \geq g_O - d_{Oi} - M(1 - w_i^A) \quad \forall i \in T \quad (52)$$

$$g_h \leq g_O - d_{Oh} + M(1 - w_h^A) \quad \forall h \in S \quad (53)$$

$$g_h \geq g_O - d_{Oh} - M(1 - w_h^A) \quad \forall h \in S \quad (54)$$

$$g_h + v_h \geq d_{hO} - M(1 - w_h^\Omega) \quad \forall h \in S \quad (55)$$

$$g_i \geq \epsilon_i + d_{iO} - M(1 - w_i^\Omega) \quad \forall i \in T \quad (56)$$

$$x_{ij}, y_{ih}, z_{hi}, w_i^A, w_h^A, w_i^\Omega, w_h^\Omega, \chi_{h,t}^-, \chi_{h,t}^+, \chi_{h,t}^{+free}, \chi_{h,t}^{-free}, C_{h,t} \in \{0, 1\} \quad \forall i, j \in T, \forall h \in S, \forall t \in [\bar{t}] \quad (57)$$

In the pricing problem, constraint (28) ensures that the route starts exactly once and ends exactly once. Constraints (29) enforce that no vehicle departs from the depot to any charging station beyond the first charging set. Flow conservation at each trip is managed by constraints (30). Flow conservation at each charging station is maintained by constraints (31). Constraints (32) restricts the route to at most one charging activity per set S_l by limiting the sum of arrivals to stations in each charging set to be no more than one, and constraints (33) defines the total number of charging activities $\bar{l}$ which is limited by L .

Time feasibility is handled by constraints (34), which ensures that if trip j follows trip i (i.e., $x_{ij} = 1$), then the end time of trip i plus the travel time d_{ij} does not exceed the start time of trip j . Constraints (35) and (36) compute the start and end times of a charging activity at station h , respectively, by accounting for the travel time from the depot (or from a preceding trip) to the station and from the station to the subsequent trip (or depot). These are given by the assumption that the vehicle should stay at the charging station for as long as possible in order for it to benefit the system by acting as batteries. Constraints (37) requires that the charging activity starts before it ends.

Constraints (38) states that, exactly one of paid charging, free charging, idle, or discharging is selected whenever a charging event occurs. Constraints (39) and (40) force the time blocks t to lie between the computed charging start time cst_h and the charging end time cet_h when a charging event occurs. Along with the objective, these three equations allow a charging event to take place only during the times the vehicle is actually at the charging station, between cst and cet . The net amount of energy charged or discharged at each station is then computed by constraints (41).

The update of the state of charge (SoC) throughout the route is managed by constraints (42)-(56).

This formulation enables flexible modeling of vehicle-to-grid scheduling decisions that are responsive to fluctuating energy conditions and grid constraints. By capturing timing feasibility, energy flow dynamics, and logical sequencing of trip and charging transitions, this MILP provides a rigorous foundation for route generation in the presence of V2G integration.

Since column generation only requires identifying columns with negative reduced cost, it is sufficient to find any feasible solution that improves the master objective. Speed up strategies such as early stopping or adding multiple columns can significantly reduce computing time [8].

8.1.2 Battery Scheduling Pricing Problem

For the stationary battery scheduling case, we again assume a uniform maximum capacity G for all batteries.

Unlike the EV setting, the batteries are assumed to be stationary; thus, the decision variables χ^+ and related terms no longer depend on the charging station h and only on the time t . With a slight abuse of notation, we reuse variables from section 8.1.1 and define χ_t^+ , χ_t^{+free} , χ_t^- , and χ_t^{-free} to carry the same meaning as the respective variables from the EVSP-V2G pricing problem. The parameter c_t denotes the price of charging at time t .

Similarly, the battery SoC variable g_t is indexed only by time, representing the available energy at the beginning of time period t . The evolution of g_t is driven exclusively by the charging or discharging decisions within each period.

The stationary battery pricing problem closely follows the structure of (27), but with the spatial routing component removed:

$$\text{Minimize } \overline{c_{\text{batt}}} + \sum_{t \in [\bar{t}]} c_t (\chi_{h,t}^+ - \chi_{h,t}^-) - \left[\sum_{t \in [\bar{t}]} \left(\beta_t^* (\chi_t^{+free} - \chi_t^{-free}) + \gamma_t^* \chi_t^- \right) \right] \quad (58)$$

s.t.

$$g_1 = G \quad (59)$$

$$g_{t+1} = g_t + \chi_t^+ + \chi_t^{+free} - \chi_t^- - \chi_t^{-free}, \quad \forall t = 1, \dots, \bar{t} - 1 \quad (60)$$

$$0 \leq g_t \leq G, \quad \forall t \in [\bar{t}] \quad (61)$$

$$\chi_t^+, \chi_t^{+free}, \chi_t^-, \chi_t^{-free} \in \{0, 1\}, \quad \forall t \in [\bar{t}] \quad (62)$$

Constraint (59) fixes the initial state of charge to the maximum capacity G , reflecting a fully charged battery at the start of the scheduling horizon. SoC update is managed by constraints (60), while constraints (61) restricts SoC to lie within physical limits $[0, G]$ for all periods, preventing infeasible overcharging or depletion. Finally, constraints (62) ensures that the activity indicators take binary values.

8.2 Additional Results and Solver Diagnostics

Table 5: Full solver diagnostics for EVSP-V2G runs ($\epsilon = 2$). Columns: *Locs* = number of task locations; *Tasks* = number of trip tasks; *LP l.b.* = LP relaxation lower bound; *MIP Obj* = MIP objective; *Run (s)* = total running time; *EV cols*, *Batt cols* = number of EV and battery columns generated; *Gap (%)* = final optimality gap.

Locs	Tasks	LP l.b.	MIP Obj	Run (s)	EV cols	Batt cols	Gap (%)
2	10	540.0	550	26	12893	3440	0.00%
3	30	1643.3	1660	53	11152	5000	1.00%
4	60	3343.3	3360	140	15694	5000	0.49%
5	100	5503.3	5540	537	15489	5872	0.66%
6	150	8330.0	8370	817	19671	11602	0.48%
7	210	11730.0	11810	2688	28097	17860	0.68%
8	280	15696.7	15840	4378	19808	9157	0.90%
9	360	20230.0	20420	14224	34614	20555	0.93%
10	450	25330.0	25510	29722	29285	16621	0.71%

Table 6: Full solver diagnostics for EVSP–V2G runs ($\epsilon = 1.5$). See Table 5 for column definitions.

Locs	Tasks	LP l.b.	MIP Obj	Run (s)	EV cols	Batt cols	Gap (%)
2	10	425.0	460	40	12629	4977	0.00%
3	30	1275.0	1290	611	35106	17384	0.88%
4	60	2550.0	2550	950	30415	9538	0.00%
5	100	4250.0	4250	3800	30167	9378	0.00%
6	150	6375.0	6430	5808	24807	10055	0.86%
7	210	8925.0	9010	13841	31825	36195	0.94%
8	280	11900.0	12030	21433	33862	15697	0.94%

References

- [1] João Almeida, Bruno Canizes, João Soares, and Zita Vale. Energy consumption and renewable generation data of 5 aggregators - 15 minute resolution (13 bus grid). Zenodo, 2020. URL <https://doi.org/10.5281/zenodo.4399670>.
- [2] Matías Alvo, Gustavo Angulo, and Mathias A Klapp. An exact solution approach for an electric bus dispatch problem. *Transportation Research Part E: Logistics and Transportation Review*, 156:102528, 2021.
- [3] Michael Brower. Electric vehicles and the case for resilience. *Clean Energy Group*, 2023. URL <https://www.cleanenergygroup.org/electric-vehicles-and-the-case-for-resilience/>.
- [4] George B. Dantzig and Philip Wolfe. Decomposition principle for linear programs. *Operations Research*, 8(1):101–111, 1960. ISSN 0030364X, 15265463. URL <http://www.jstor.org/stable/167547>.
- [5] Marelot de Vos and Rolf Nelson van Lieshout. Electric vehicle scheduling with capacitated charging stations and partial charging. *arXiv preprint arXiv:2207.13734*, 2022.
- [6] Guy Desaulniers, Fausto Errico, Stefan Irnich, and Michael Schneider. Exact algorithms for electric vehicle-routing problems with time windows. *Operations Research*, 64(6):1388–1405, 2016.
- [7] Guy Desaulniers, Fabio Errico, Stefan Irnich, and Michael Schneider. Electric vehicle scheduling and optimal charging problem: Complexity, exact and heuristic approaches. *ResearchGate*, 2024. URL https://www.researchgate.net/publication/278829014_Electric_Vehicle_Scheduling_and_Optimal_C
- [8] Jacques Desrosiers, Marco Lübbecke, Guy Desaulniers, and Jean Bertrand Gauthier. Branch-and-price. Les Cahiers du GERAD G-2024-36, Groupe d’études et de recherche en analyse des décisions, GERAD, Montréal QC H3T 2A7, Canada, October 2024. URL <https://www.gerad.ca/en/papers/G-2024-36>.
- [9] R Gilmore and Ralph Gomory. A linear programming approach to the cutting stock problem i. *Oper Res*, 9, 01 1961. doi: 10.1287/opre.9.6.849.
- [10] Andrew Holland and Caitlin Joyce. Microgrids for the 21st century: The case for a defense energy architecture. *National Defense University Press*, 2023. URL <https://ndupress.ndu.edu/Media/News/News-Article-View/Article/3678506/microgrids-for-the-21>
- [11] Ideal Energy. Building energy resiliency for the military with microgrids. *Ideal Energy Solar*, 2019. URL <https://www.idealenergysolar.com/building-energy-resiliency-for-the-military-with-microgrid>
- [12] Maroš Janovec and Michal Koháni. Exact approach to the electric bus fleet scheduling. *Transportation Research Procedia*, 40:1380–1387, 2019.
- [13] Dawid Jóźwiak, Jayakrishnan R. Pillai, Pavani Ponnaganti, and Jan Jantzen. Optimal charging and discharging strategies for electric cars in pv-bess-based marina energy systems. *Electronics*, 12(4):1033, 2023. doi: 10.3390/electronics12041033.

- [14] L.V. Kantorovich and V.A. Zalgaller. *Calculation of Rational Cutting of Stock*. Lenizdat, Leningrad, 1951. In Russian.
- [15] Mert Keskin and Bülent Çatay. Partial recharge strategies for the electric vehicle routing problem with time windows. *Transportation Research Part C: Emerging Technologies*, 65:111–127, 2016.
- [16] Lu Li, Hong K Lo, and Feng Xiao. Mixed bus fleet scheduling under range and refueling constraints. *Transportation Research Part C: Emerging Technologies*, 104:443–462, 2019.
- [17] Andrea Marr and Wilson Rickerson. Generating security: Resilient renewable power for u.s. military installations. *OurEnergyPolicy.org*, 2014. URL <https://www.ourenergypolicy.org/wp-content/uploads/2014/04/Generating-Security-Resilient-Renewable-Power-for-U.S.-Military-Installations.pdf>
- [18] Abul Masrur, Annette G. Skowronska, Janie Hancock, Steven W. Kolhoff, and et al. Military-based vehicle-to-grid (v2g) and vehicle-to-vehicle (v2v) microgrid – system architecture and implementation. *IEEE Transactions on Transportation Electrification*, 3(3):xx–xx, 2017. doi: 10.1109/TTE.2017.2779268.
- [19] Gordon Maynard, Pride Metcalf, and Andrew Baughman. San nicolas island uesc energy conservation projects. Presented at the Federal Utility Partnership Working Group Seminar, San Diego, CA, May 2, 2019, 2019. URL <https://www.energy.gov/sites/prod/files/2019/05/f62/11-fupwg-spring-2019-maynard.pdf>.
- [20] Ann-Sophie Pepin, Guy Desaulniers, Alain Hertz, and Dennis Huisman. A comparison of five heuristics for the multiple depot vehicle scheduling problem. *Journal of scheduling*, 12:17–30, 2009.
- [21] Shyam SG Perumal, Richard M Lusby, and Jesper Larsen. Electric bus planning & scheduling: A review of related problems and methodologies. *European Journal of Operational Research*, 301(2):395–413, 2022.
- [22] Marco Rinaldi, Erika Picarelli, Andrea D’Ariano, and Francesco Viti. Mixed-fleet single-terminal bus scheduling problem: Modelling, solution scheme and potential applications. *Omega*, 96:102070, 2020.
- [23] Ons Sassi and Ammar Oulamara. Joint scheduling and optimal charging of electric vehicles problem. In *Computational Science and Its Applications–ICCSA 2014: 14th International Conference, Guimarães, Portugal, June 30–July 3, 2014, Proceedings, Part II 14*, pages 76–91. Springer, 2014.
- [24] Maximilian Schiffer and Günther Walther. Electric vehicle routing: A literature review of models, methods and future research directions. *European Journal of Operational Research*, 261(3):815–828, 2017.
- [25] Michael Schneider, Andreas Stenger, and Dominik Goeke. The electric vehicle-routing problem with time windows and recharging stations. *Transportation science*, 48(4):500–520, 2014.
- [26] Mostafa Sedighizadeh, Seyed Saeed Fazlhashemi, H. Javadi, and M. Taghvaei. Multi-objective day-ahead energy management of a microgrid considering responsive loads and uncertainty of the electric vehicles. *Journal of Cleaner Production*, 267:121562, 2020. doi: 10.1016/j.jclepro.2020.121562.

- [27] Xindi Tang, Xi Lin, and Fang He. Robust scheduling strategies of electric buses under stochastic traffic conditions. *Transportation Research Part C: Emerging Technologies*, 105:163–182, 2019.
- [28] Marcel E van Kooten Niekerk, JM Van den Akker, and JA Hoogeveen. Scheduling electric vehicles. *Public Transport*, 9:155–176, 2017.
- [29] Weitiao Wu, Yue Lin, Ronghui Liu, and Wenzhou Jin. The multi-depot electric vehicle scheduling problem with power grid characteristics. *Transportation Research Part B: Methodological*, 155:322–347, 2022. ISSN 0191-2615. doi: <https://doi.org/10.1016/j.trb.2021.11.007>. URL <https://www.sciencedirect.com/science/article/pii/S0191261521002125>.
- [30] Nancy A. Youssef and Gordon Lubold. U.s. tackles a military vulnerability in the pacific: Supply lines. *The Wall Street Journal*, 2024. URL <https://www.wsj.com/world/u-s-tackles-a-military-vulnerability-in-the-pacific-supply-lines->
- [31] Le Zhang, Shuaian Wang, and Xiaobo Qu. Optimal electric bus fleet scheduling considering battery degradation and non-linear charging profile. *Transportation Research Part E: Logistics and Transportation Review*, 154:102445, 2021.